Ziwen Hua

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College Park, MD

Education

University of Maryland, College Park, MD, USA 09/2026 - 06/2030 (Expected)
Undergraduate Student, Major Undeclared

Shanghai No.2 High School, Meilong Campus, Shanghai, China 09/2022 - 06/2025
High School Diploma, 2025

Internship & Experience

Terps Racing IC, Electronics Subteam Member 09/2026 - Present

- Designed Brake System Plausibility Device (BSPD) circuit boards for TR27

Shanghai HintonAI Tech, Intern 07/2026 - 08/2026

- Designed and improved the hand-eye calibration module for a ROS2-based open source robotic painting system, RobRoss.
- Collaborated with the path-execution pipeline to ensure calibration outputs were robust in real robot runs, reducing coordinate drift and alignment errors during autonomous drawing.

Research & Projects

Research on Preparation of Rapid Cooling Equipment 09/2023 - 06/2024
Shanghai Jiao Tong University

- Studied college physics and industrial cooling technology, and studied methods for physics research.
- Produced a mini semiconductor refrigerator in the lab and wrote a paper.

Application of Generative Pre-trained Transformer Models 09/2023 - 06/2024
Independent Research Project, Shanghai No.2 High School

- Researched the use of modern AI models in daily life, learned Python programming and knowledge of probability and statistics.
- Trained a small AI model to translate Chinese into English.

Skills & Hobbies

- **Programming:** Python, C, Java; MATLAB/Simulink for modeling and simulation
- **Robotics:** ROS2; hand-eye calibration and coordinate-frame transforms for robot arms
- **Electronics & CAD:** KiCad schematic and PCB layout (BSPD for Formula SAE); SolidWorks part and assembly modeling
- **Machine Learning:** Core modern AI algorithms and the full model training and inference workflow
- **Tools:** Git/GitHub, LaTeX, Linux command line
- **Languages:** Chinese (native), English (fluent)
- **Hobbies:** Photography, Racing